Smart Accident Detection & Alert System

Using ESP8266 + IMU (MPU6050)

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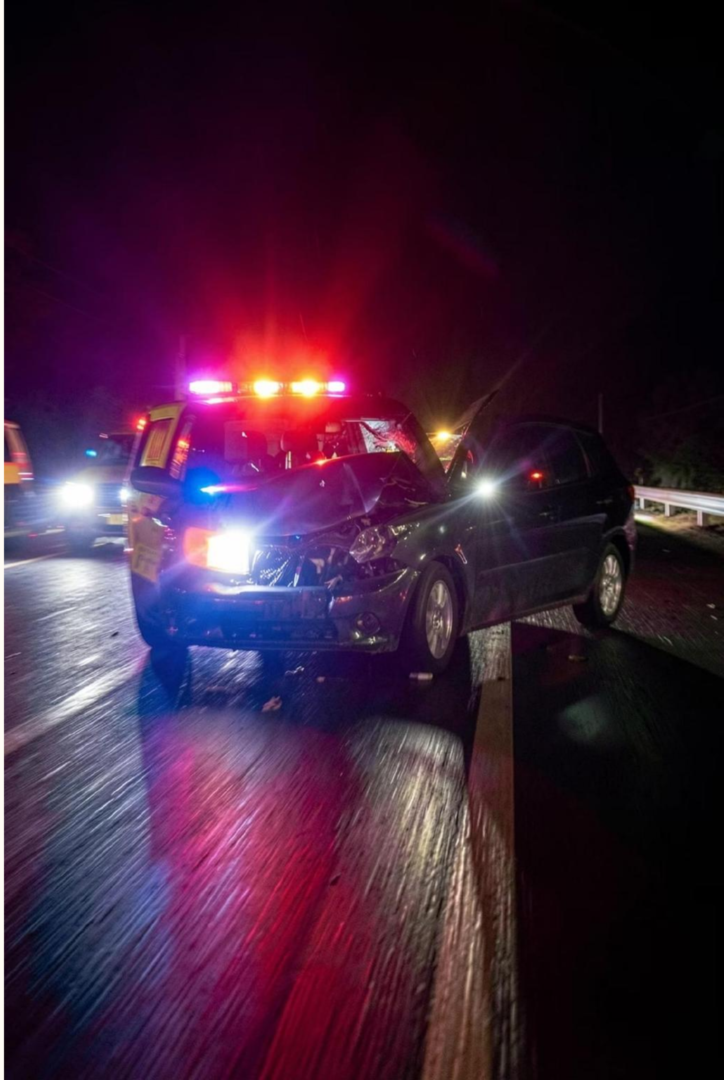
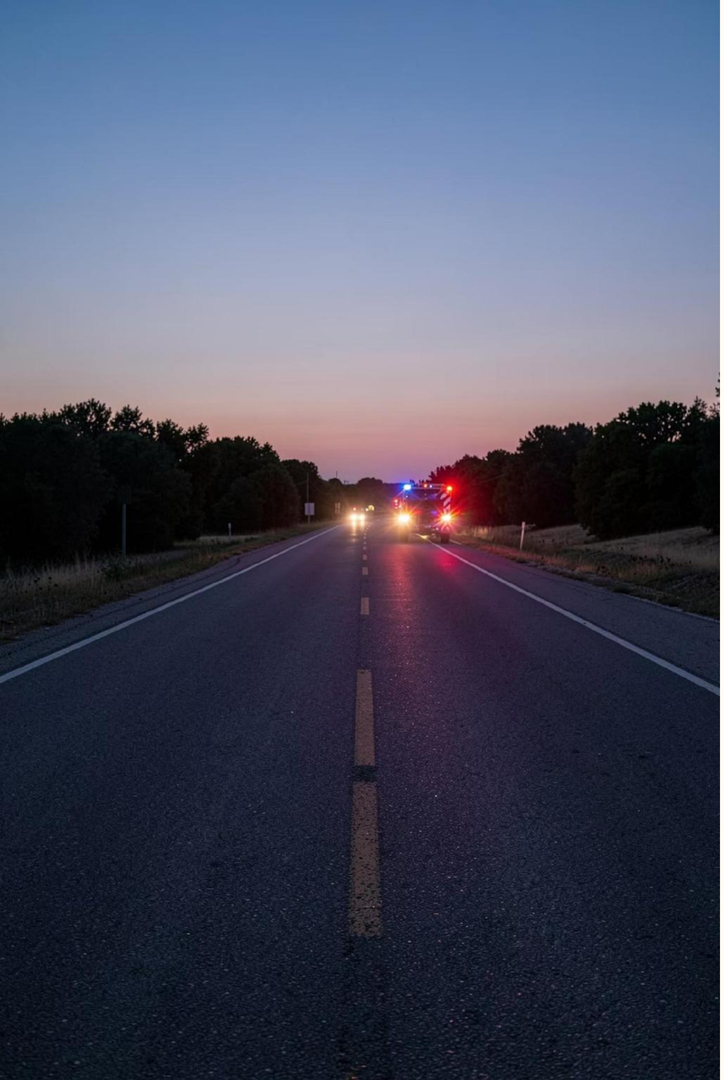




Why This Project?

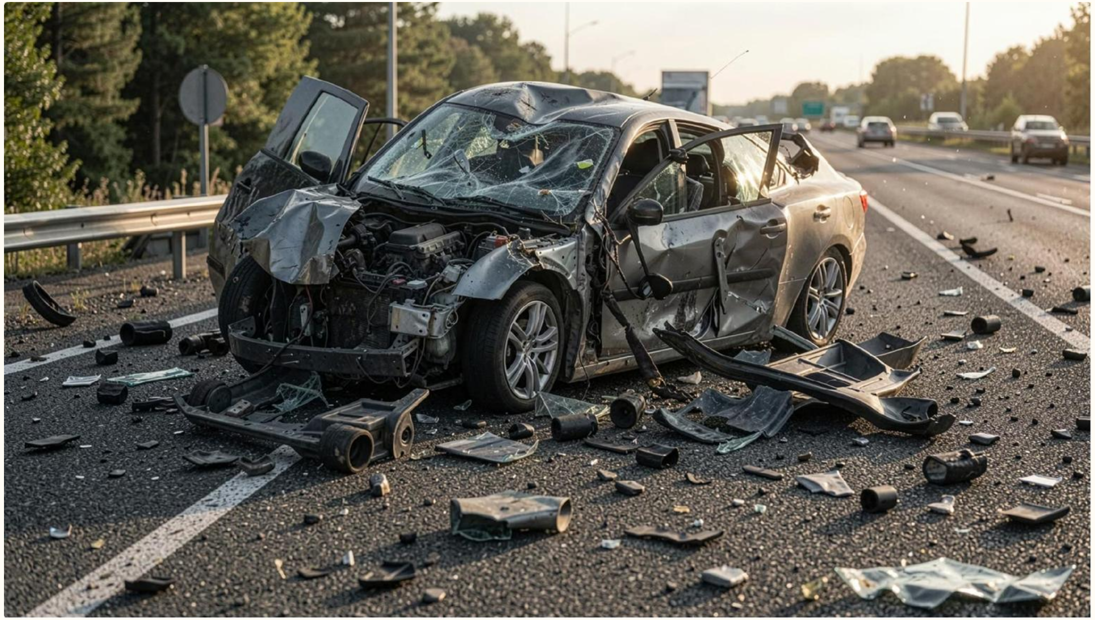
The Problem

Road accidents happen suddenly. Many victims do not receive help in time, and delays in emergency response can worsen injuries.

Our Goal

To build a low-cost, smart safety system that automatically detects accidents and sends real-time alerts using Wi-Fi — no human intervention needed.

Problem Statement



What Exactly Goes Wrong?

- Accidents occur due to overspeeding, sudden impact, or vehicle rollover — often without warning.
- People nearby may not realise an accident has happened immediately.
- Emergency services are delayed, increasing the risk of serious injury or death.

Who Suffers Because of This?

Drivers & Passengers

Directly affected by impact; need immediate medical attention.

Families of Victims

Left without information or support during critical moments.

Emergency Teams

Delayed alerts reduce their ability to respond effectively.

Society

Road accidents cause loss of life, injuries, and economic burden.

Our Solution

SMART ACCIDENT DETECTION SYSTEM

The system uses an **ESP8266 (NodeMCU)** microcontroller paired with an **MPU6050 IMU sensor** to continuously monitor vehicle motion. When a sudden impact is detected, the system automatically triggers a Wi-Fi alert — no manual input required.

🔍 Detect

IMU monitors acceleration in real time

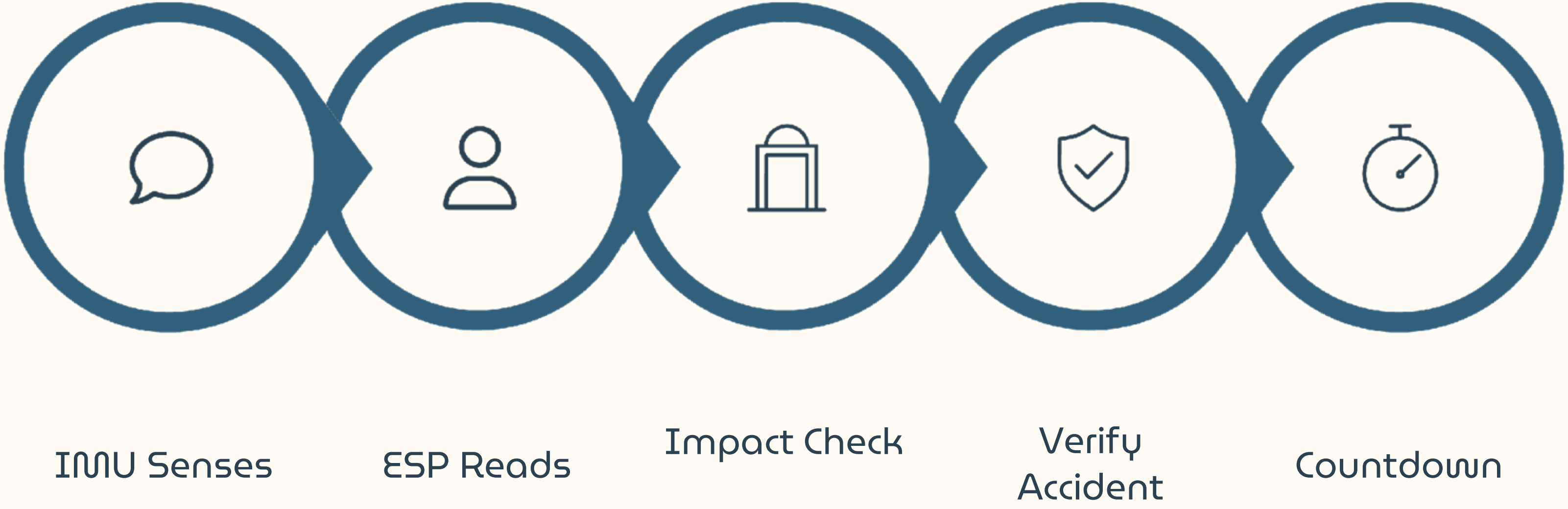
⚡ Process

ESP8266 analyses sensor data instantly

📡 Alert

Wi-Fi notification sent automatically

How It Works



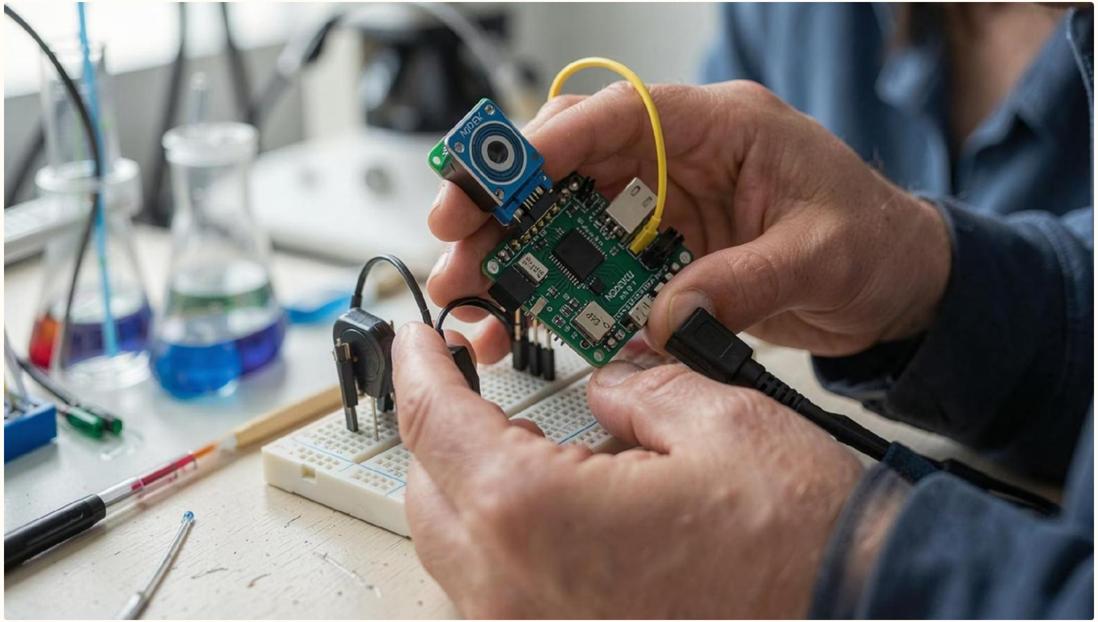
The countdown step is a key safety feature — it allows the driver to cancel a false alert before the notification is sent, ensuring the system is both smart and reliable.

Components Used

Component	Purpose
ESP8266 (NodeMCU)	Wi-Fi connectivity and data processing
MPU6050 IMU Sensor	Measures acceleration and rotation
Buzzer	Audible warning sound
Push Button	Cancel false alert during countdown
Breadboard & Jumper wires	Circuit connections and prototyping

All components are low-cost and widely available — making this system easy to replicate and scale.

Our Working Model



01

Show the ESP8266 board and MPU6050 sensor connections.

02

Demonstrate accident detection by safely shaking or tilting the model.

03

Show the countdown timer activating on the serial monitor.

04

Display the final Wi-Fi alert message being sent.

Why This Solution Helps

Faster Reporting

Accidents are reported within seconds — no waiting for a bystander to call for help.

Low Cost

Built with affordable, off-the-shelf components accessible to students and developers.

Easy to Install

Compact design fits in cars, bikes, and industrial vehicles with minimal setup.

Reduces Delay

Faster alerts mean emergency teams can respond sooner, potentially saving lives.

Future Scope

Where Can This Project Go?

Add **GPS** for live location sharing with emergency services.

Send alerts directly to **family members and hospitals**.

Store accident data in the **cloud** for analysis.

Use **AI/ML** for more accurate, adaptive detection.

Integrate with **ambulance services** and smart city systems.

Thank You!

We hope this project inspires others to use technology for real-world safety challenges.

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